

Mobile Phone Sales Performance Analysis

Project Overview

This Power BI project provides an end-to-end data analytics solution built to evaluate the sales trends and operational throughput of a mobile retail and distribution business in India. The dashboard visualizes sales across major smartphone brands (Apple, Samsung, OnePlus, Vivo, Motorola), customer ratings, payment preferences, and geographic distribution.

Problem Statement

Mobile retail stakeholders often struggle to monitor sales KPIs across multiple brands, regional stores, and fast-shifting payment modes.

- **The Goal:** Build a centralized dashboard to track **Total Sales (769M)**, **Transactions (4K)**, **Total Quantity Sold (19K)**, and **Average Sales Value (40K)**.
- **Key Challenges Solved:** Identifying regional sales hotspots in India, tracking monthly demand fluctuations, mapping customer rating structures, and identifying which payment gateways (UPI, Credit Card, Debit Card, Cash) are used most.

Power Query ETL & Data Cleaning

Before visualizing, the raw transaction dataset (Sales_Data) underwent systematic cleaning steps starting at the 5:40:00 mark of the master class:

1. **Data Type Correction:** Ensuring Total Sales and Price Per Unit were set to fixed currency, while Transaction ID and date keys were correctly formatted to prevent mathematical errors.
2. **Missing/Null Check:** Filtering out null fields in the Brand and Mobile Model columns to protect chart accuracy.
3. **Date Attribute Extraction:** Breaking down raw date strings to create a dynamic sidebar filter tracking months (January to December) and calculating day names.

DAX Formulation & Measures

To populate the top KPI cards and dynamic charts, the following core DAX measures were written inside the Sales_Data table:

- **Total Sales Revenue:**

Code snippet

```
Total Sales = SUM(Sales_Data[Sales_Amount])
```

- **Total Transactions Count:**

Code snippet

```
Transactions = COUNT(Sales_Data[Transaction_ID])
```

- **Total Quantity Transacted:**

Code snippet

Total Quantity = SUM(Sales_Data[Units_Sold])

- **Average Ticket Value:**

Code snippet

Average = AVERAGE(Sales_Data[Sales_Amount])



Dashboard UI Layout & Visual Insights

- **Left Sidebar Navigation:** A clean button panel mapping calendar months seamlessly to filter the entire report with one click.
 - **Geographic Map Visual:** Tracks Total Sales by City across India (cities like Mumbai, Lucknow, Patna, Bhopal, Hyderabad), pointing out where demand is strongest.
 - **Monthly Quantity Trend:** A line chart displaying seasonal fluctuations (showing peaks like July at 1,700 units).
 - **Payment Distribution Pie Chart:** Highlights transaction channels, clearly mapping out the market share between UPI, Debit Card, Cash, and Credit Cards.
 - **Funnel Chart for Ratings:** Aggregates Customer Ratings (from 1 to 5 stars) to show brand satisfaction levels clearly.
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